# Boon Depths - Technical Sheet

## Tech used

One self-contained HTML file (about 29 KB), no libraries, no external assets. HTML5 Canvas 2D with additive blending and glow, scaled to the device pixel ratio (max 2x) in a 360x560 portrait playfield. Web Audio API for all sound: synthesized effects and a procedural synthwave track that speeds up in the boss fight. localStorage for meta progress. Touch, mouse and keyboard input; haptic pulses where supported.

## Systems

Pure game core (BD module) separated from rendering so it can be simulated headless. 12 chambers with scaling enemy count, HP (+20%/chamber) and damage (+12%/chamber); 4 enemy types plus a 2-phase boss. 12 gifts across 4 patrons, up to level 3 each. Springs of Rest at chambers 4 and 8. Auto-aim arc strike, i-frame dash, radial Flare. Death defiance. Ember economy and 4-track permanent upgrade shop. Auto-pause when the app goes to the background.

## Audit 1: Balance (simulated play)

A bot played 60 runs each at three upgrade levels. First version: 97% of fresh characters won in 1.5 minutes, far too easy, and runs paid 300 embers. Fixes: steeper enemy scaling, more and faster foes, boss HP 900 to 1800, ember rewards roughly halved. Result: fresh 10% win (average chamber 9.8), mid upgrades 67%, fully upgraded 98%, about 135 to 270 embers per run, so the full Mirror takes roughly 6 to 8 runs.

## Audit 2: Bugs

Headless harness ran full runs through gifts, springs, pausing, death and the menu with 0 errors. Fixed: the room-intro and tutorial text used total run time instead of chamber time, so they only showed in chamber 1; a button could stay held if the OS cancelled a touch (added touchcancel); a leftover no-op in the music scheduler was removed.

## Audit 3: Player experience

Added a first-run control hint, a CHAMBER N / THE WARDEN banner on every room, a red low-HP pulse, cooldown fills on all three buttons, a boss health bar, floating damage numbers and CRIT callouts, screenshake and vibration on hits, and auto-pause when the app is backgrounded.

## Testing note

Tested with simulated play and a headless harness, not yet on a physical phone.